Index Reconstitutions and Stock Returns in Sweden

Price responses, fund holdings and the limits of inference

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Abstract

We examine stock returns and benchmark-linked fund holdings around OMX Stockholm Benchmark reconstitutions from July 2007 through June 2026. Adapting the empirical questions of Greenwood and Sammon (2025), we distinguish a decline in average event-window returns from evidence that those returns are economically negligible. The complete return sample contains 256 additions and 215 deletions. Additions earn an average market-adjusted cumulative return of 8.52% in 2007–2011 and 1.50% in 2022–2026. The latest-period 95% wild-bootstrap interval includes zero, but is too wide to establish disappearance. Across 2017–2026, the addition return is 2.15%, with an interval of [1.01%, 3.39%]. Return responses coexist with more concentrated trading before implementation. Fund-holdings evidence is consistent with benchmark-related demand, but changes in reporting coverage and uncertainty in the demand denominator weaken comparisons of price-to-demand ratios. A new exploratory, fixed-cohort calculation finds a positive cumulative addition return through twenty trading days after implementation in the later decade. Neither that persistence nor the other mechanism diagnostics identifies a causal or permanent effect of index membership. The contribution is a documented Swedish adaptation that separates return patterns, measurement choices and identifying assumptions.

Keywords: index reconstitution; event studies; passive investing; stock demand; clustered inference.

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Research scope: OMXSB. Proposed pension-procurement analyses in earlier project plans are outside this study; no such outcomes are reported.

1 Introduction

Index reconstitutions create an unusually observable intersection between portfolio mandates and securities trading. A membership announcement identifies the securities entering or leaving a benchmark and a date on which the change becomes effective. For investors tracking that benchmark, the announcement changes a portfolio target. For other investors, it reveals potential future trading demand. The resulting price path depends on how anticipated that demand is, which investors absorb it, and whether the announcement also conveys information about the firms selected. An index event is therefore informative about the relation between demand and prices, but it is not automatically an exogenous demand experiment.

The question has acquired renewed importance because a declining return response need not imply declining passive investment. Greenwood and Sammon (2025) document that the S&P 500 addition return fell from 7.4% in the 1990s to less than 1% in the later sample, even as assets linked to the index increased. Their published article also reports an average deletion return of approximately 0.1% in 2010–2020. Those findings motivate a distinction between the amount of benchmark-related demand and the price movement associated with a particular reconstitution. Changes in one need not map proportionally into changes in the other.

This paper asks whether an analogous decline is visible in Sweden and, more narrowly, whether recent OMX Stockholm Benchmark (OMXSB) returns support the claim that the index effect has disappeared. We use "index effect" as shorthand for a measured event-window return, not as a causal designation. The primary outcome is the sum of a security's daily return minus the OMXSB gross total-return index return from one trading day before the announcement through one trading day after implementation. We complement this outcome with holdings changes, trading-volume concentration, composition adjustment, comparison securities and within-firm share-class contrasts.

Sweden provides a useful setting because a rules-based benchmark permits a close examination of the relation between selection, anticipation and implementation. Its selection rules nevertheless combine liquidity and industry representation. A simple market-capitalization threshold does not determine membership. In addition, benchmark variants and mixed-fund mandates make the relevant investor population broader than funds carrying an exact OMXSB label. These features motivate a measurement exercise that connects returns to dated fund classifications while retaining a clear boundary between observed holdings changes and exogenous order flow.

The documented event list contains 536 membership changes across 39 review cycles. The complete main return sample contains 256 additions and 215 deletions, after return-data requirements and ten deletion exclusions associated with corporate events. Effective dates run from July 2007 through June 2026; the last year is incomplete. Because many securities change membership at the same review, event counts overstate the number of independent sources of information. Our main uncertainty estimates therefore cluster by review cycle and use a restricted wild cluster bootstrap. We also examine equal-review weighting, cluster jackknife uncertainty, and dependence across repeated observations of the same security.

Three findings organize the analysis. First, average addition returns are lower in the

final five-year block than in the first, but the path is not a steady decline. Total-window returns are 8.52%, 0.25%, 2.85% and 1.50% across the four successive blocks. The latest-minus-earliest contrast is -7.03 percentage points, with a pointwise bootstrap interval excluding zero. The later-decade mean, however, remains 2.15%, with a 95% interval of [1.01%, 3.39%]. Deletion returns are less precisely estimated. Their later-decade mean is -1.04%, with an interval spanning both zero and economically material negative returns.

Second, the inference changes when the question changes. The most recent addition estimate has a 95% interval of approximately [-0.10%, 3.22%]. Failure to reject zero is consequently compatible with positive returns that would be economically meaningful. An explicitly labeled practical-equivalence check cannot place the latest-period mean inside a band of plus or minus one percentage point. The recent-period conclusion also depends on influential events and simultaneous MSCI changes. The wider 2017–2026 result is more stable across those exclusions. We therefore distinguish attenuation relative to the early sample from disappearance in a recent sample.

Third, measured trading demand does not provide a clean causal explanation for the remaining returns. In the common later-decade sample of 136 additions with usable benchmark-linked holdings data, mean sign-adjusted CAR is 2.71% and measured net holdings growth is 1.03% of shares outstanding. Their ratio is 2.63. This ratio describes a particular covered sample; it is neither an instrumental-variable estimate nor an identified demand elasticity. In sparse early samples, jointly accounting for uncertainty in the numerator and denominator produces very wide or unbounded confidence sets. Apparent precision in some inherited percentile intervals is therefore not sufficient support for strong claims about changes in market elasticity.

The mechanism evidence is informative in a more limited sense. Trading becomes more concentrated on the day immediately preceding effectiveness: the average share of a 45-trading-day volume window on that day rises from 9.44% to 15.24% for additions. Daily data cannot establish that the additional concentration takes place in the closing auction, and concentration need not imply additional total trading. Quoted spreads are lower in later years, but their association with returns also reflects common time variation. The share-class exercise yields an imprecise relative return and cannot rule out demand spillovers between classes. These findings narrow interpretations without selecting a unique explanation.

An exploratory extension makes the return-horizon question more precise. We hold the event cohort fixed and divide the period into non-overlapping windows before and after implementation. For 169 additions in 2017–2026, the average cumulative return from AD-1 through ED+20 is 3.30%; it remains positive in separate exclusions of Sivers and simultaneous MSCI events. This calculation is a new analysis of already observed data, not an independent confirmation. It establishes a finite-horizon association. It does not establish permanent price impact, because later fundamental news, selection and benchmark-model error can also generate a persistent cumulative return.

The paper's contribution is thus a transparent empirical boundary. The Swedish data support positive later-decade returns around additions, coexistence with concentrated implementation trading, and substantial uncertainty about demand-scaled effects. They do not establish that index inclusion causes those returns, that improved liquidity explains their

evolution, or that a recent insignificant coefficient demonstrates disappearance. All central estimates are traceable to a frozen data version. An independent calculation reproduces the inherited return arithmetic, while a separate audit records numerical and source limitations instead of equating reproducibility with validity.

The remainder of the paper positions the study in the nearest literature, describes the institutional setting and sample, specifies the estimands and inference, and presents the return and mechanism evidence. Robustness analysis and the discussion then separate conclusions supported by the estimates from conclusions requiring a stronger design.

2 Related literature

Our closest reference is Greenwood and Sammon (2025). We adapt its questions about the evolution of index-event returns and the relation between those returns, benchmark-linked demand, composition and trading conditions. This is a Swedish empirical adaptation, rather than a numerical replication of the US sample. The principal extension is the explicit treatment of changing coverage, common-sample ratios and weak denominator inference in a smaller, clustered event sample. The paper also reports a fixed-cohort finite-horizon calculation. It does not claim to reproduce every test in the anchor study.

Earlier index research motivates several interpretations of the same price pattern. Shleifer (1986) links index additions to downward-sloping demand for stocks, while Harris and Gurel (1986) emphasizes price pressure and subsequent reversal. Chang et al. (2015) studies index assignment using a regression discontinuity design. The last comparison is particularly important here: our size-screened comparison securities do not establish a comparable assignment discontinuity. A coefficient that survives observable adjustment remains an association unless the selection assumptions are defensible.

Chinco and Sammon (2024) shows why named index funds and the broader population of passive investors need not coincide. This motivates careful labeling of our covered benchmark-linked fund population, rather than interpreting it as the entire passive sector. It also makes zero measured holdings growth an ambiguous observation when reporting coverage is incomplete.

The statistical design draws on event-study and clustered-inference research. MacKinlay (1997) provides the event-study framework; Petersen (2009) and Cameron et al. (2011) motivate attention to dependence across time and securities. MacKinnon et al. (2023) explains why clustering alone need not deliver reliable inference with few or influential clusters. We use Fieller (1954) to distinguish an imprecise ratio from a well-identified parameter, and Holm (1979) to show the effect of a limited multiplicity correction. These methods improve the reporting of uncertainty; they do not supply the missing economic identification.

3 Institutional setting and data

3.1 Index construction and event timing

OMXSB is designed to represent a selection of large and actively traded securities listed in Stockholm. The current methodology combines turnover ranking with free-float capitalization within industry supersectors. High-turnover securities qualify for inclusion, while the least liquid portion of the universe is ineligible; the industry selection targets at least 85% of each supersector's free-float capitalization, subject to the eligibility rules. The index is free-float capitalization weighted. This combination of rules makes membership related to firm size, past trading and industry composition (Nasdaq, 2026).

The current review calendar uses reference dates at the end of April and October, announcements after the close on the sixth trading day before effectiveness, and implementation at the opening of the first trading day in June or December. Historical event dates are taken from the event-specific source list, not retroactively assigned from this current calendar. The sample includes earlier January and July implementations. The earlier rules document provides a historical institutional reference, but neither that document nor the current methodology substitutes for a notice-by-notice event calendar (NASDAQ OMX, 2013; Nasdaq, 2026).

Selection rules also change within the sample. The May 2012 notice explicitly announces revised industry-diversification criteria based on the Industry Classification Benchmark, alongside 11 additions and 24 removals (NASDAQ OMX, 2012). Historical period comparisons therefore span institutional changes as well as changes in investor demand and market conditions. The unusually large review should not be treated as evidence that the assignment process was constant over time. This documented change motivates sensitivity to review influence; it does not supply an exogenous break design.

We denote the recorded announcement trading date by AD and the effective trading date by ED. If implementation occurs at the opening of ED, the previous trading day is the final trading day under the old composition. This distinction is important for the volume analysis. A large volume share at ED-1 is consistent with portfolio adjustment before effectiveness, but daily volume does not identify auction volume. Announcement timestamps also matter. The symmetric three-day announcement window includes the next trading day when information is released after the close; daily data still cannot isolate a narrow announcement-time response.

Membership selection need not be independent of future returns. Past price increases can raise capitalization, past trading can affect liquidity eligibility, and corporate developments can influence both eligibility and expected cash flows. A predictable rule therefore does not by itself make membership changes as good as random. The economic interpretation below explicitly allows selection and information effects to accompany trading demand.

3.2 Sample construction and missing observations

The frozen event file contains 284 additions and 252 deletions. Each row includes the company, security identifiers, event type, review cycle, announcement and effective dates,

and a source URL. The imported price-event panel contains 487 candidates. Its reduction from 536 reflects the available historical security mappings and return data, rather than an economic exclusion of all missing firms. Missing observations are not assumed to be random.

Table 1: Construction of the return sample

Stage	Additions	Deletions
Documented membership changes	284	252
Imported return candidates	261	226
After deletion exclusions	261	216
Complete main return window	256	215

The source list contains 536 membership changes in 39 review cycles from 2007 through the first review of 2026. The imported return panel contains 487 candidates. Ten deletions flagged for offers or delisting are excluded; six remaining candidates lack a complete main window. The final sample contains 471 event observations. Each event type spans 38 review cycles; their union spans 39. Source: frozen event files and independent reconstruction, audit/results/sample_flow.csv.

The main deletion sample excludes ten candidates flagged for an offer, merger-related event, or delisting under the inherited screening rule. Other deletions flagged only for changes in share counts remain in the return analysis but may be excluded from holdings-based calculations. Requiring a complete main return window leaves 256 additions and 215 deletions. The total sample spans 39 review cycles, although each event type is observed in only 38 of them. The asymmetry is why type-specific cluster counts must be reported rather than inferred from the pooled sample.

We divide events by effective year into 2007–2011, 2012–2016, 2017–2021 and 2022–2026. These are calendar blocks of equal nominal length, not equally informative samples: the study begins in July 2007 and stops after the June 2026 review. Event counts, coverage and review sizes also differ. Pooling the first two and last two blocks produces the earlier and later decades. The period labels should not be read as evidence that the endpoints were independently selected before the research process. The repository records earlier period definitions and subsequent revisions.

Selection has two distinct consequences. It can change the mean return when omitted securities differ systematically from included securities. It can also change the precision of that mean through the size and balance of review clusters. Later complete return samples are larger, but their availability does not repair missing historical observations. We report sample counts at every material step and avoid inferring an economic time trend directly from the growth in usable observations.

3.3 Returns, prices and trading data

Daily security data are supplied through LSEG. The principal return file includes closing prices, daily total returns, volume, market capitalization, shares outstanding and closing bid and ask quotes. Separate index files supply OMXSBGI and other market series. Broader stock panels support the liquidity and comparison exercises. The data are licensed research

inputs; a public replication package can identify them and document transformations without redistributing the source observations.

The inherited daily return construction uses the reported total return when available and falls back to the percentage change in the closing price otherwise. It requires every abnormal-return observation within an event window to be present. The independent audit reproduces the announcement, implementation and total-window CARs for all 487 imported candidates. Seven candidates use at least one price-return fallback within the main window, covering 104 event-day observations. Eliminating implicit forward filling in that fallback does not change a main-window CAR in this frozen sample. This check validates the implemented arithmetic; it does not establish that every vendor observation or historical identifier is correct.

OMXSBGI is a gross total-return benchmark, so the benchmark and reported stock total returns are aligned more closely than a price-only index would permit. Nevertheless, a market-adjusted return imposes a unit market exposure and zero intercept. Differences in risk exposure, sector performance, dividends in fallback observations and changes in benchmark membership can remain in the residual. We therefore call the outcome market-adjusted CAR and retain the benchmark as part of its definition.

3.4 Fund holdings and the demand sample

The holdings analysis uses LSEG/Lipper fund positions around the events and a dated classification assembled from fund mandates and source documents. Three nested groups are retained: B contains strict OMXSB followers; V adds benchmark variants; C adds relevant mixed-fund exposure. The broad C measure is the main descriptive demand proxy, while B and V reveal how the measurement changes with the investor definition. An index-trade diagnostic checks the frequency of purchases of additions and disposals of deletions. The production code reads a dated mandate list; it does not mechanically reclassify every fund-year from that diagnostic alone.

Eligible fund-event comparisons require a report within 120 calendar days before AD and a report from ED through 45 calendar days afterwards, together with a stable classification across the two reporting dates. The code avoids treating a stale pre-event holding returned as an "after" observation as a new position. It nevertheless infers some zero positions from the presence of a report elsewhere in the observed fund data. That is a coverage assumption: the absence of a security row need not always mean a true zero if an underlying report or vendor extract is incomplete.

The change in reported shares held is normalized by the event's shares-outstanding denominator and sign-adjusted for deletions. Events with flagged splits are excluded from the principal ratio estimates to reduce inconsistencies between adjusted holdings and outstanding shares. The measurement still requires historical identifiers and share-class denominators to be correct. The longer interval between holdings reports also contains trading unrelated to the index event. Accordingly, the holdings change is not an observed implementation-day order or an independently assigned demand shock.

The resulting sample is much smaller than the return sample. The first addition block has eight usable C observations across five reviews, while the final block has 78 across nine

reviews. The later decade contains 136 usable additions and 108 deletions. Comparing the full-sample CAR with demand measured only for those subsets would confound differences in price response with differences in coverage. Every reported ratio instead uses the same events for its numerator and denominator.

3.5 Other measurements and provenance

The study also uses MSCI membership-change records, a targeted company-news screen, a set of size-screened comparison securities, and share-class pairs. The MSCI exercise asks whether removing overlapping index events changes the OMXSB return pattern. The news screen covers the 24 most extreme inherited returns. It is not a comprehensive news audit and cannot certify the remaining sample as free of firm information. The share-class data include supplemental price sources, which motivates a separate analysis excluding source changes.

All reported estimates refer to one frozen repository version. Forty-three input files pass their recorded checksum tests. The archived code is run separately from the frozen output, and the independent return reconstruction does not import its return or inference functions. Source URLs and retrieval outcomes are preserved in the audit package. Complete notice reconciliation and report-level holdings checks remain unresolved; the paper does not recast an inherited verification flag as a new independent source verification.

4 Empirical design and inference

4.1 The estimand and its causal boundary

Let r_{it} denote security i's daily return and r_{mt} the OMXSBGI return, both expressed in percentage points. The market-adjusted cumulative return over a trading-day window $[a, b]$ is

$$CAR_i[a, b] = \sum_{t=a}^{b} (r_{it} - r_{mt}). \tag{1}$$

The primary window is $[AD_i - 1, ED_i + 1]$. Short announcement and implementation windows are $[AD_i - 1, AD_i + 1]$ and $[ED_i - 1, ED_i + 1]$. These quantities are arithmetic sums of daily excess returns, not compounded buy-and-hold abnormal returns. The short-window means need not add to the primary-window mean because the primary window also includes intervening days, and because completeness requirements can select different observations.

For event type s and period p, the principal parameter is the mean of this measured CAR among usable events of that type and period. It is not the average treatment effect of index membership. To identify such an effect, the benchmark-adjusted counterfactual return would have to remove relevant common movements, selection into membership would need to be conditionally unrelated to the remaining counterfactual outcome, and coincident information would need to be excluded or accounted for. The present design does not establish those conditions.

An illustrative decomposition clarifies the problem. Suppose an observed event return can be written as

$$CAR_i = \lambda_i Q_i + I_i + B_i + U_i, \tag{2}$$

where Q_i is unexpected benchmark-related demand, λ_i its price response, I_i is contemporaneous information, B_i is selection-related expected return, and U_i collects remaining benchmark and measurement error. Equation (2) is an interpretive accounting device, not an estimated structural model. A change in mean CAR can reflect any component, including covariance between demand and its price response. Neither index membership nor a positive holdings change separately identifies λ_i.

This framework yields discriminating questions rather than mutually exclusive mechanism tests. A stronger pre-announcement return is consistent with anticipation but also with selection on earlier performance. Lower quoted spreads are consistent with cheaper trading but do not identify the supply curve faced by benchmark traders. Positive post-event returns can reflect incomplete reversal, continued information or a misspecified benchmark. We evaluate these implications while retaining the alternatives.

4.2 Period contrasts and observable composition

The unadjusted specification regresses CAR on four mutually exclusive period indicators. Each coefficient is the event-weighted period mean. Linear contrasts test the difference between the earliest and latest blocks, the last two blocks, and the two decades. A separate regression on calendar year summarizes a linear trend without imposing the four bins. The trend is a descriptive projection over an uneven historical sample, not evidence of a constant structural decline.

The composition specification adds centered measures of abnormal turnover, relative market capitalization, residual return variance and analyst coverage. It is estimated separately for additions and deletions. The inherited variable construction uses information near AD, and some variables include AD-1, which also lies in the outcome window. This timing overlap is another reason to interpret the exercise as an adjustment rather than a clean conditioning strategy for causal identification. We report an unadjusted regression on the identical complete-case sample beside the adjusted regression. A change between full-sample and adjusted estimates otherwise mixes sample selection with covariate adjustment.

The comparison-security regression adds review fixed effects, log capitalization and a pre-announcement return measure to an event indicator. Nonmember comparison securities for additions pass a size screen based on the smallest addition; member comparisons for deletions pass the corresponding screen based on the largest deletion. These screens are not the Nasdaq assignment rule. They neither demonstrate local random assignment nor guarantee comparable latent fundamentals. The comparison coefficient is therefore a conditional return difference. Moreover, the inherited pre-return control includes AD-1 and overlaps the outcome by one day. The regression remains a secondary descriptive diagnostic until that overlap and historical eligibility construction are independently rebuilt.

4.3 Demand scaled returns

Let $s_i = 1$ for additions and $s_i = -1$ for deletions. We sign-adjust both returns and holdings changes so that positive values correspond to index-directed movements. For a covered fund group g, define

$$D_{ig} = 100 s_i \frac{\sum_{f \in g} \Delta H_{if}}{S_i}, \qquad Y_i = s_i CAR_i[AD_i - 1, ED_i + 1], \qquad \widehat{M}_{pg} = \frac{\overline{Y}_{pg}}{\overline{D}_{pg}}, \quad (3)$$

where ΔH_{if} is the change in covered fund f's reported shares and S_i is shares outstanding. Both averages use the same non-split complete cases. The ratio measures percentage points of sign-adjusted CAR per percentage point of measured ownership change. It is a ratio of means, not a mean of event-level ratios.

The distinction matters statistically and economically. A small mean denominator can make the ratio large even when its numerator is imprecise. If the covered holdings measure is $D^{obs} = \kappa Q + \eta$, where κ captures coverage and η measurement error or unrelated trading, the ratio is not invariant to changes in coverage. Even with negligible η, a changing κ changes the observed ratio without a change in the underlying price response. Fund assets, disclosed holdings and actual implementation orders are therefore not interchangeable denominator measures.

The inherited analysis reports percentile intervals from resampling review clusters. We retain those outputs in the audit package but emphasize a Fieller sensitivity calculation that uses the joint covariance of the two means. A candidate ratio m is retained when

$$(\overline{Y} - m\overline{D})^2 \leq t^2_{G-1,0.975} \left[\widehat{V}_{YY} - 2m\widehat{V}_{YD} + m^2 \widehat{V}_{DD} \right]. \quad (4)$$

Here $\widehat{V}$ is the review-clustered CR1 covariance matrix of $(\overline{Y}, \overline{D})$. The set can be bounded, consist of two unbounded rays, or include every real value. Such outcomes are information about weak denominator precision, not computational failures to be discarded. With very few reviews, the t approximation is itself uncertain; the Fieller analysis is an informative sensitivity calculation rather than a claim of exact coverage (Fieller, 1954).

4.4 Clustering and finite sample uncertainty

The main return tables use the inherited restricted wild cluster bootstrap. For each null hypothesis, the code imposes the coefficient restriction, multiplies residuals by a common random weight within each review, re-estimates the model and compares studentized statistics. It uses 9,999 draws, Rademacher weights when at least 12 clusters are available and Webb weights otherwise. Confidence intervals are obtained by numerical test inversion. The review is the natural first clustering unit because several securities share an announcement and implementation cycle.

This procedure does not make reviews independent by construction. A firm can reappear in multiple reviews, macroeconomic conditions can persist, and a period comparison may have only a few influential reviews behind one side of the contrast. We therefore supplement the main results with review jackknife uncertainty and review-by-security

two-way clustering, following the dependence concerns discussed by Petersen (2009); Cameron et al. (2011); MacKinnon et al. (2023). Equal-review means answer a different weighting question and are reported as such. Leave-one-review-out ranges show influence; they are not confidence intervals.

The rerun audit also reveals a numerical limitation in some auxiliary bootstrap inversions. The original routine searches outward from the point estimate, stops at a finite search boundary, and returns a single interval. In sparse designs, that need not characterize a potentially irregular acceptance set. Floating-point differences can change some of those endpoints substantially. We do not rely on unstable early-period auxiliary ratio intervals. Composition-adjustment intervals in the manuscript use analytical CR1 uncertainty, explicitly labeled, and the ratio table reports Equation (4). Main return-bootstrap intervals remain the frozen pointwise results, whose rerun differences are small and documented.

4.5 Multiplicity, power and economic disappearance

The research record contains many periods, windows, screens and earlier exploratory analyses. It would be inaccurate to describe all reported tests as fixed before the data were observed. We treat the inherited analysis as exploratory and avoid selecting the preferred specification by its significance. As a limited diagnostic, Appendix C applies Holm adjustment to a declared family of ten inherited return tests. This does not account for all historical researcher choices and cannot retroactively create a confirmatory design (Holm, 1979).

The archived minimum-detectable-effect calculation is

$$MDE \approx \left(t_{G-1,0.975} + t_{G-1,0.80}\right) \widehat{SE}. \tag{5}$$

This is a plug-in analytical approximation for a two-sided 5% test at 80% power. It is not the exact power of the implemented wild bootstrap, nor does it account for the full specification search. We use it to describe the scale of differences that the sample can resolve, rather than calculating "observed power" as a new argument for the estimated effect.

Disappearance requires more than failure to reject zero. For a transparent audit benchmark, we ask whether a 90% analytical confidence interval for the latest-period mean lies wholly within $[-1, 1]$ percentage points, corresponding to two one-sided 5% equivalence tests. The margin is chosen for reporting after the old results were known, not as a preregistered or universally valid economic threshold. Failure of this check means that this dataset does not establish practical equivalence at that margin; it does not prove that the true effect exceeds it.

4.6 A fixed cohort horizon extension

The new extension divides the cumulative return into non-overlapping intervals:

$$CAR_i[AD_i - 1, ED_i + 20] = CAR_i[AD_i - 1, ED_i - 1] + CAR_i[ED_i, ED_i + 20]. \tag{6}$$

All three quantities are calculated on the same complete-case cohort. This avoids combining a short-window mean with a post-event mean based on different securities, or counting ED and ED+1 twice. We report all four five-year blocks and both event types in the machine-readable audit, with the later decade and latest block emphasized in the main table.

The extension uses independently coded analytical CR1 intervals and a separate restricted wild-bootstrap test of the mean with 49,999 draws. Simulated p-values use $(k+1)/(B+1)$, where k is the number of exceedances. The bootstrap p-value and analytical interval are deliberately labeled as different summaries. The extension was specified after inherited outcomes were known; additional exclusions were examined after viewing the new results. A nonzero cumulative mean rejects a particular zero-mean finite-horizon hypothesis under the maintained return model. It does not identify an index-induced permanent level shift.

5 Return patterns around reconstitution

5.1 Lower recent averages without a monotonic disappearance

Table 2 reports the announcement, implementation and total-window return means. Additions have positive total-window means in every period, but their magnitudes vary sharply. The first block is exceptional at 8.52%. The second falls to 0.25%, the third rises to 2.85%, and the latest is 1.50%. This sequence is compatible with an attenuation from the unusually high early observations, but it is not a monotonic decline toward zero.

Table 2: Market adjusted returns around index changes

Type	Period	N	G	AD	ED	Total	95% interval	p
Addition	2007–2011	33	9	0.96	1.16	8.52	[2.46, 14.74]	0.027
Addition	2012–2016	54	10	1.87	-0.73	0.25	[-3.73, 5.30]	0.908
Addition	2017–2021	82	10	-0.16	0.38	2.85	[1.26, 5.20]	0.005
Addition	2022–2026	87	9	1.20	-0.09	1.50	[-0.10, 3.22]	0.062
Deletion	2007–2011	33	9	-0.77	-0.09	-6.16	[-11.41, -2.10]	0.007
Deletion	2012–2016	46	10	-1.71	-0.85	-2.47	[-14.42, 3.60]	0.522
Deletion	2017–2021	61	10	-2.09	0.89	-0.43	[-3.43, 3.02]	0.759
Deletion	2022–2026	75	9	-0.71	-1.23	-1.53	[-4.24, 0.89]	0.279

Returns are percentages. AD denotes CAR $[AD-1, AD+1]$, ED denotes CAR $[ED-1, ED+1]$, and Total denotes CAR $[AD-1, ED+1]$, where AD is the recorded announcement trading date and ED the effective trading date. The benchmark is OMXSBGI. Means weight events equally. N and G are event and review counts for Total; AD and ED use their own complete cases. Intervals and two-sided p-values for Total use the inherited restricted wild cluster bootstrap with 9,999 draws, Webb weights below 12 clusters and Rademacher weights otherwise. These are pointwise, exploratory intervals. No stars are used. Source: effect/tabell1_lang.csv, panel A/B.

Announcement-window means are substantially smaller than the first period's total-window mean. Implementation-window means are close to zero in the latest block: -0.09% for additions and -1.23% for deletions, with wide intervals in the underlying outputs. A small implementation-window return does not imply an absence of benchmark-related

trading. If investors anticipate the reconstitution, or if liquidity supply absorbs trading near implementation, any associated price movement may occur elsewhere. Conversely, a return between announcement and implementation can also reflect unrelated information over that interval.

Deletion means are -6.16%, -2.47%, -0.43% and -1.53% across the four blocks. Their uncertainty is large, especially in 2012–2016. In that block, the event-weighted mean and equal-review mean differ considerably, illustrating why a larger event count cannot replace balanced independent review information. The economically relevant statement is that deletion returns are less precisely resolved, rather than that the deletion channel is absent.

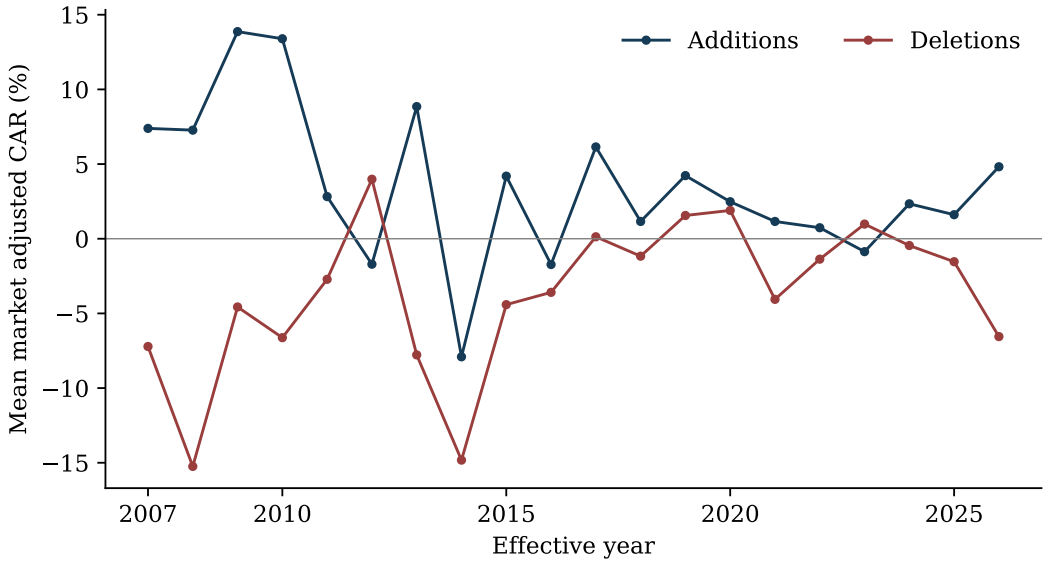


Figure 1: Annual mean market-adjusted returns. Each point is the event-weighted CAR $[AD - 1, ED + 1]$ for additions or deletions in an effective calendar year. The final point covers only the first 2026 review. Lines connect descriptive annual averages and are not fitted trends. Some years contain very few usable events; the figure carries no annual inferential bands. Source: effect/effect_by_year.csv.

Table 3 separates three different time-comparison questions. The latest-minus-earliest addition contrast is -7.03 percentage points, with a pointwise bootstrap interval of [-13.10, -1.10]. The corresponding deletion contrast is +4.62 percentage points: deletions become less negative relative to the first block. Both contrasts concern the endpoints of this sample. Neither establishes a gradual annual decline, and their nominal significance weakens under the ten-test audit correction.

The change from 2017–2021 to 2022–2026 is much less precisely estimated. For additions it is -1.35 percentage points, with an interval spanning zero. The later-minus-earlier decade contrast is -1.24 percentage points and is also imprecise. The linear addition trend is approximately -0.28 percentage points per year, again with a bootstrap interval including zero. These estimates do not contradict the large earliest-to-latest difference; they show that the shape and timing of the attenuation are unresolved by a single summary coefficient.

The repository contains a previously selected breakpoint around June 2010. Because

Table 3: Changes in the event window return

Type	Contrast	Estimate	95% interval	p
Addition	Latest minus earliest	-7.03	[-13.10, -1.10]	0.029
Addition	Latest minus preceding	-1.35	[-3.88, 0.98]	0.250
Addition	Later minus earlier decade	-1.24	[-5.26, 2.54]	0.502
Addition	Linear trend per year	-0.28	[-0.64, 0.08]	0.117
Deletion	Latest minus earliest	4.62	[0.25, 9.81]	0.039
Deletion	Latest minus preceding	-1.10	[-5.11, 2.70]	0.564
Deletion	Later minus earlier decade	2.97	[-2.94, 8.60]	0.328
Deletion	Linear trend per year	0.19	[-0.29, 0.64]	0.400

The outcome is CAR $[AD - 1, ED + 1]$. Differences are percentage points; the linear trend is percentage points per calendar year. Earliest, preceding, and latest periods are 2007–2011, 2017–2021, and 2022–2026. Decades are 2007–2016 and 2017–2026. Restricted wild bootstrap inference clusters by review cycle, using 9,999 draws. A difference of significance levels across periods is not a test of a difference. Source: effect/tabell1_lang.csv, panel D.

that timing emerged in earlier analysis of the same return history, a conventional test that subsequently treats it as fixed understates the uncertainty associated with searching for the break. We do not promote that breakpoint to a new identified regime change. The annual plot and the full set of period contrasts provide the descriptive evidence without implying a verified event behind the apparent change.

5.2 Recent returns and the meaning of an insignificant estimate

Pooling 2017–2026 yields a mean addition CAR of 2.15%, with a 95% wild-bootstrap interval of [1.01%, 3.39%]. This average covers 169 additions over 19 review cycles. The corresponding deletion mean is -1.04%, with an interval of approximately [-2.88%, 0.90%], over 136 deletions and 19 cycles. The sign and precision differ, but we do not infer an addition-deletion asymmetry from separate significance decisions; such an asymmetry would require a direct contrast on a consistent estimand.

The final five-year addition block contains 87 observations over nine reviews. Its 1.50% estimate has a bootstrap interval spanning zero and returns above 3%. The archived analytical MDE is approximately 2.35 percentage points. These facts make a null result weak evidence for disappearance. At the audit's one-percentage-point margin, the 90% analytical interval is [0.13%, 2.87%], so it does not lie inside the equivalence band. Deletions also fail equivalence, with an interval of [-3.62%, 0.55%].

The interval for the latest addition mean cannot be interpreted as a confidence interval for a causal membership premium. It is uncertainty about the market-adjusted mean among the observed events under the maintained inference model. The economic interpretation is narrower but still useful: the recent sample cannot distinguish a small effect from a positive association of meaningful magnitude. A statement that the effect has "disappeared" would omit precisely the uncertainty that matters for that question.

5.3 Composition and comparison securities

Table 4 reports the observable-composition adjustment. For additions, the latest-period estimate is 1.59% without controls on the common sample and 1.60% with controls. The early-period values are 8.97% and 10.12%, respectively. In this particular linear adjustment, the four covariates do not explain away the early-to-late difference. That finding is descriptive: the variables may not capture the relevant selection process, their construction uses nearby price information, and some are themselves related to anticipated membership.

Table 4: Composition adjustment on a common event sample

Type	Period	Unadjusted	Adjusted	Adjusted CR1 interval
Addition	2007–11	8.97	10.12	[5.09, 15.16]
Addition	2012–16	0.74	0.55	[-3.01, 4.11]
Addition	2017–21	2.64	2.33	[0.38, 4.29]
Addition	2022–26	1.59	1.60	[0.68, 2.53]
Deletion	2007–11	-5.66	-6.23	[-11.03, -1.43]
Deletion	2012–16	-1.37	-1.42	[-7.61, 4.76]
Deletion	2017–21	-0.77	-0.29	[-3.35, 2.77]
Deletion	2022–26	-1.83	-1.95	[-4.02, 0.12]

Period intercepts from CAR $[AD - 1, ED + 1]$ regressions. Both columns use the same complete-case sample: 237 additions or 185 deletions. Adjusted estimates control for centered abnormal turnover, relative size, residual return variance and analyst coverage. The columns are not a causal decomposition. Intervals are analytical CR1 review-clustered intervals using t_{G-1}; unstable inverted bootstrap intervals from auxiliary sparse specifications are not used here. Source: composition/tabell2_tabell4.csv, table 2.

For deletions, the common-sample latest-period estimates are -1.83% without controls and -1.95% with controls. Differences between these numbers and the full return table partly reflect complete-case selection. The adjusted regression contains 185 deletions, compared with 215 in the main return table. The separate common-sample column prevents that reduction from being attributed entirely to economic controls.

The size-screened comparison regression offers another descriptive perspective. The later-decade addition coefficient is 1.53 percentage points, with a bootstrap interval of [0.36, 2.68]. Restricting comparison securities to the main market produces 1.39 percentage points, with an interval of [0.12, 2.60]. The corresponding later-decade deletion coefficient is -1.40 percentage points, with an interval reaching slightly above zero. Table 5 reports the main comparison results and counts.

These results show that the later addition association is not mechanically identical to the average return of the available comparison group. They do not show that a particular fraction of the raw CAR is caused by index membership. The raw and conditional estimates use different samples, weights and assumptions. Subtracting one from the other would not isolate a causal selection component. The lack of a reconstructed Nasdaq assignment boundary and the overlapping pre-return control remain substantive limitations.

Table 5: Returns relative to size screened comparison securities

Type	Period	Events	Controls	Coefficient	95% interval	p
Addition	2007–2016	78	787	3.90	[0.10, 8.07]	0.046
Addition	2017–2026	159	1609	1.53	[0.36, 2.68]	0.015
Deletion	2007–2016	68	415	-3.24	[-9.27, 1.17]	0.210
Deletion	2017–2026	130	833	-1.40	[-2.75, 0.03]	0.053

The coefficient is on an event indicator in a regression of CAR $[AD - 1, ED + 1]$ on that indicator, pre-announcement returns, log market capitalization and review fixed effects. Comparison observations are security-review observations, not unique firms. The inherited size screens do not reconstruct the Nasdaq inclusion threshold and are not a regression discontinuity design. The pre-return control includes AD-1, which also enters the outcome; this overlap is an additional reason to treat the coefficients as descriptive adjustments. Intervals and p-values use 9,999 restricted wild cluster draws by review. Source: near_additions/near_additions_results.csv.

6 Holdings, trading and mechanism evidence

6.1 Benchmark linked holdings and the denominator problem

Table 6 relates returns to reported holdings changes on a common sample. For additions, mean C-group purchases rise from 0.15% of shares outstanding in 2007–2011 to 1.11% in 2017–2021, before declining to 0.97% in 2022–2026. These are changes in observed fund ownership, not estimates of the total amount of passive capital tracking the index. The first comparison is particularly fragile because it involves only eight addition events across five reviews. Figure 2 displays the annual series without implying that its changing coverage is economically neutral.

The early addition ratio is 57.87 percentage points of CAR per percentage point of covered ownership change. It would be tempting to compare this with 2.35 in the latest block and describe an extraordinary increase in the elasticity of stock supply. That conclusion is not warranted. The early denominator is small, the numerator is imprecise, and the observed investor set expands over time. The 95% Fieller set for the early ratio runs from -9.55 to 471.88. Even its sign is not precisely resolved by this calculation. The earliest deletion sample is weaker still: seven observations across three reviews produce a Fieller set containing every real value.

For later additions, the holdings measurement is better supported. Pooling 2017–2026 yields a common-sample return of 2.71%, a C-group ownership change of 1.03%, and a ratio of 2.63. Its Fieller interval is [0.91, 4.37]. This supports a positive association between the measured numerator and a positive measured demand denominator in the later pooled sample. It does not validate the ratio as a structural inverse elasticity. Some of the measured trading may respond to the same information that moves prices, and the reporting interval is broader than the return window.

The period-specific inference is less decisive. The 2017–2021 and 2022–2026 addition Fieller sets both include zero, even though their inherited percentile-bootstrap intervals do not. The distinction arises because the procedures summarize joint numerator-denominator uncertainty differently. Neither is an automatic gold standard with nine or ten clusters.

Table 6: Returns scaled by observed benchmark linked fund purchases

Type	Period	N	G	$\bar{Y}$	$\bar{D}_C$	$\widehat{M}$	Fieller set
Addition	2007–11	8	5	8.78	0.15	57.87	[-9.55, 471.88]
Addition	2012–16	27	8	-1.94	0.52	-3.76	[-15.09, 4.24]
Addition	2017–21	58	10	3.29	1.11	2.97	[-0.14, 6.30]
Addition	2022–26	78	9	2.28	0.97	2.35	[-0.21, 4.73]
Addition	2017–26	136	19	2.71	1.03	2.63	[0.91, 4.37]
Deletion	2007–11	7	3	-0.19	0.26	-0.72	All real values
Deletion	2012–16	25	7	2.07	0.48	4.36	[-17.73, 30.17]
Deletion	2017–21	40	10	0.20	0.99	0.20	[-4.05, 4.89]
Deletion	2022–26	68	9	1.66	1.18	1.41	[-0.87, 2.83]
Deletion	2017–26	108	19	1.12	1.11	1.01	[-0.88, 2.61]

$Y = s_i CAR_i[AD - 1, ED + 1]$ and D_C is the sign-adjusted change in covered benchmark-linked funds' holdings as a percentage of shares outstanding; $s_i = 1$ for additions and -1 for deletions. Positive values therefore indicate movement in the index-directed direction for both event types. $\widehat{M} = \bar{Y}/\bar{D}_C$ uses the same non-split complete-case sample for both means. Group C includes strict followers, benchmark variants, and benchmark-linked mixed-fund holdings. The last column is a 95% Fieller confidence set using the joint CR1 covariance of both means and t_{G-1}, an asymptotic sensitivity measure rather than exact small-sample inference. It differs from the inherited percentile bootstrap. Source: fund_demand/omrakning_resultat.csv, table 5, and audit/results/fieller_ratios.csv.

Reporting the discrepancy is more informative than selecting the narrower interval. For deletions, the later-decade ratio is 1.01 and its Fieller interval includes zero. Sign adjustment changes the direction convention; it does not make addition and deletion mechanisms symmetric.

An independent reporting channel provides a useful measurement comparison. In the matched 2019–2026 addition sample with 106 events across 15 reviews, the Finansinspektionen-based ownership measure is 1.14%, compared with 1.05% for the matched C measure. The common-sample CAR is 2.43%, giving ratios of 2.12 and 2.30, respectively. Their inherited bootstrap intervals are [0.87, 3.79] and [0.93, 4.15]. Agreement on this overlap is reassuring about scale for the observed later sample. It cannot establish full fund coverage, recover the missing early ownership denominator, or make holdings changes exogenous.

The economic conclusion from the holdings evidence is consequently limited but substantive. Benchmark-linked investors demonstrably change their reported positions around these events, and their observed trades are large enough that ignoring demand would be uninformative. The evidence does not tell us what fraction of the return is generated by those trades. Historical comparisons of price-to-demand ratios require comparable investor coverage, correctly adjusted share counts, and a credible separation of unexpected index demand from other trading. Those conditions remain research requirements rather than established facts.

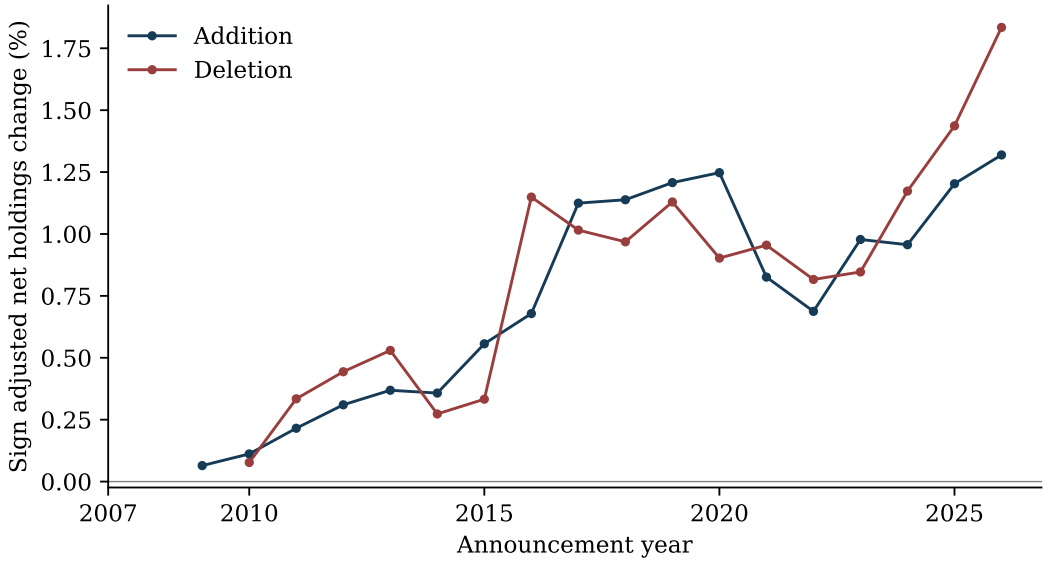


Figure 2: Measured ownership changes around index events. Annual means of sign-adjusted changes in C-group holdings as a percentage of shares outstanding. Positive deletion values indicate net sales. The series uses the holdings sample; coverage changes over time and 2026 is incomplete. It is neither total index-tracking assets nor implementation-day trading. Source: fund_demand/figur1_skikt.csv.

6.2 Trading concentrates before implementation

Daily volume provides a different view of implementation. For each event, the inherited measure divides daily trading volume by total volume over ED-22 through ED+22, a 45-trading-day window. The final trading day before effectiveness accounts for 9.44% of that volume for early additions and 15.24% for latest-period additions. The change is 5.79 percentage points, with an inherited review-bootstrap interval of [3.30, 8.84]. Deletions show a similar rise, from 8.75% to 14.52%, a difference of 5.76 percentage points with an interval of [3.26, 8.38]. Figure 3 displays the surrounding daily profile.

This concentration and the recent small implementation-window return are not contradictory. A larger share of activity can occur on a particular day while the equilibrium price adjustment remains small. Investors may trade against anticipated demand; the supply of immediacy may have improved; or prices may already incorporate information about likely membership. The data do not separately identify those explanations. In particular, a volume share is not total turnover, and it cannot measure the execution costs paid by a benchmark fund. A smaller denominator elsewhere in the window could also increase the share.

The analysis is descriptive in two further respects. Events are averaged without equalizing their investment capacity or economic importance. The treatment of missing volume as zero can affect concentration if missingness changes over time. Reconstructing complete intraday trading and auction data would be necessary to distinguish a true shift toward closing-auction execution from changes in daily reporting and trading outside the event day.

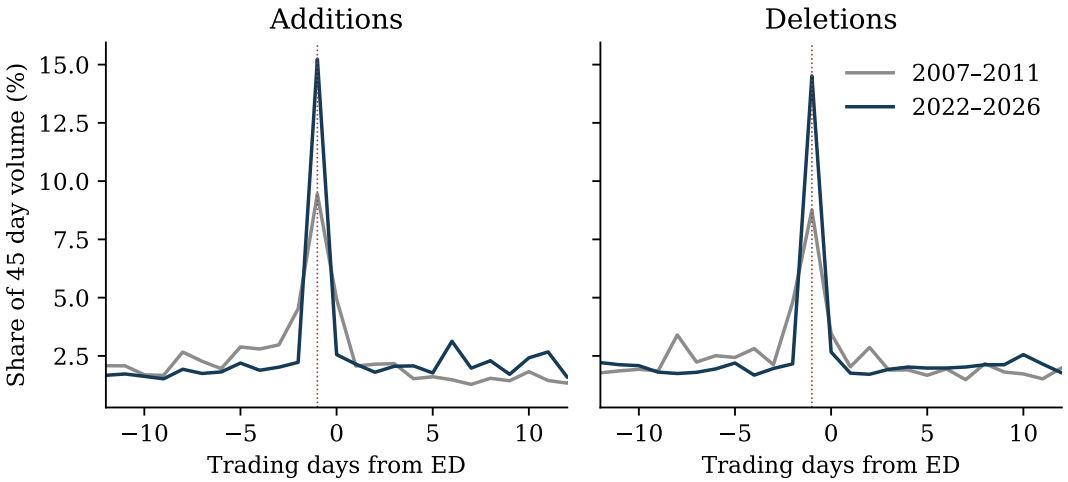


Figure 3: Concentration of trading around effectiveness. Average daily shares of each event's total volume over ED-22 to ED+22. The figure shows ED-12 to ED+12 from that fixed denominator. The dotted line marks ED-1. Missing volume is set to zero in the inherited calculation; volume-sample counts therefore differ from complete return counts. Daily observations do not distinguish auctions from continuous trading. Source: liquidity/volume_share_by_day.csv.

6.3 Quoted spreads and observable liquidity

The broader market panel shows a decline in the value-weighted closing quoted spread of OMXSB members, from 0.270% in 2007 to 0.079% in 2025. The partial 2026 observation is 0.087%. The measure is the closing ask-minus-bid spread relative to the quote midpoint; quotes above 20% are removed in the inherited construction. Figure 4 describes the evolution without attributing it to index trading.

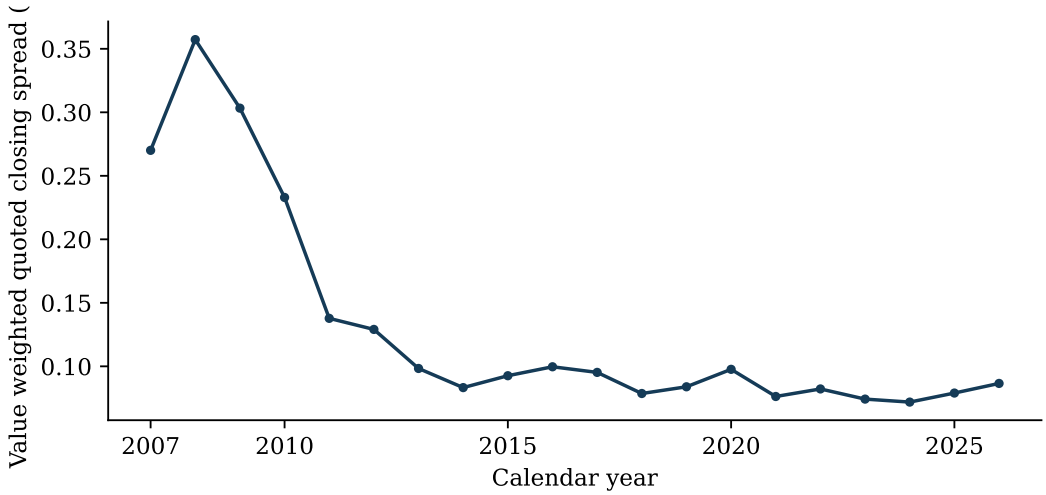


Figure 4: Quoted closing spreads among index members. Calendar-year averages of the value-weighted quoted bid-ask spread relative to the quote midpoint, in percent. The 2026 observation is incomplete. This is an observable quote measure, not an effective spread, implementation shortfall or causal measure of liquidity provision. Source: liquidity/spread_by_year.csv.

Lower quoted spreads are consistent with a market in which smaller trades can be executed more cheaply. They do not directly measure the slope of residual stock supply for a large index-related order. That slope depends on market depth, inventory capacity and the response of other investors, none of which is fully captured by the closing spread. The weighting also uses the available capitalization measure; multiple listed share classes and changing membership can affect the aggregate. The figure therefore gives institutional context for the return results, rather than a test that liquidity improvements caused their attenuation.

Taken together, holdings and volume indicate that small implementation-window returns do not mean index-related activity has become unimportant. The main uncertainty is about how much of the observed return reflects that activity, when any adjustment occurs, and how the relevant demand is measured. These distinctions motivate the robustness and horizon analyses rather than a single mechanism conclusion.

7 Robustness and an exploratory horizon extension

7.1 Overlapping events, news and influential observations

Table 7 shows how the later addition means respond to separate exclusions. Removing overlapping MSCI changes reduces the 2017–2026 mean from 2.15% to 1.46%, with an interval of [0.50%, 2.47%]. The latest-period mean falls from 1.50% to 0.73%, with an interval including zero. Overlap events have unusually large average returns: 11.25% among the 12 later-decade overlaps and 14.16% among the five latest-block overlaps. This makes the overlap consequential for the magnitude of the OMXSB association.

Table 7: Sensitivity of addition returns

Period	Specification	N	CAR	95% interval
2017–26	Baseline	169	2.15	[1.01, 3.39]
2017–26	Without MSCI overlap	157	1.46	[0.50, 2.47]
2017–26	Without screened company news	163	1.91	[1.01, 2.82]
2017–26	Winsorized 1st and 99th percentiles	169	1.92	[0.91, 3.05]
2017–26	Without Sivers	168	1.71	[0.57, 2.96]
2022–26	Baseline	87	1.50	[-0.10, 3.22]
2022–26	Without MSCI overlap	82	0.73	[-0.77, 2.32]
2022–26	Without screened company news	84	1.30	[0.12, 2.43]
2022–26	Winsorized 1st and 99th percentiles	87	1.04	[-0.11, 2.25]
2022–26	Without Sivers	86	0.62	[-0.85, 2.23]

Event-weighted CAR $[AD-1, ED+1]$ in percent, with pointwise restricted wild cluster bootstrap intervals by review. Each exclusion is applied separately. The company-news screen examines the 24 most extreme inherited events, rather than the full event universe, and is outcome-informed. MSCI exclusion changes the target population. Winsorization changes the outcome. These are sensitivity descriptions, not independent replications or specification selection rules. Sources: effect, msci_overlap, news_screen and extreme_values aggregate CSV files.

The exclusion changes the target population. It does not identify the incremental

contribution of each index when both change. Overlapping events can differ in size, visibility, investor demand and information. The smaller non-overlap estimate is thus evidence of sensitivity to a recognizable subgroup, not a causal decomposition into OMXSB and MSCI effects. A clean attribution would require a design that distinguishes the assignment and timing of the two index changes.

The targeted news exclusion leaves a later-decade mean of 1.91% and a latest-block mean of 1.30%. These values should not be labeled news-free returns. Only the 24 most extreme inherited events were reviewed under the screen, so unexamined ordinary observations may also contain firm news. Conditioning the audit on extreme outcomes can be useful for detecting obvious problems, but it is unsuitable as a comprehensive rule for declaring all remaining returns attributable to membership.

Removing Sivers reduces the later-decade mean to 1.71%, while the latest-block mean falls to 0.62%. The former remains positive with a pointwise interval excluding zero; the latter does not. Winsorization gives the same qualitative distinction between a more stable pooled later result and a sensitive final-period estimate. Because exclusions and winsorization answer different questions, their agreement cannot be counted as independent evidence. The full-sample result remains the baseline, and the range of estimates describes its dependence on data handling.

These checks imply two useful constraints on the argument. First, the positive later-decade addition pattern is not entirely generated by one extreme stock or by the identified overlapping index events. Second, the latest five-year average is not sufficiently stable to support a confident claim about the current size of a membership premium. Neither statement licenses an unconditional claim about the population of all omitted historical events.

7.2 Dependence, weighting and repeated securities

The later addition mean of 2.15% becomes 2.50% when reviews receive equal rather than event-count weights. Removing one review at a time produces event-weighted means between 1.87% and 2.47%. The analytical CR1 standard error is 0.55 percentage points, the review jackknife standard error is also approximately 0.55, and the review-by-security two-way estimate is 0.47. In this sample, these alternatives do not reveal a hidden collapse of precision behind the later pooled mean. They still rely on assumptions about dependence and should not be interpreted as proofs that the effective independent sample size is known.

For the latest addition block, the event and equal-review means are both approximately 1.50%. The leave-one-review-out range is 1.02% to 2.01%, and analytical standard errors range from about 0.71 to 0.77 across the reported estimators. These figures place the wide latest-period interval in context: even without a uniquely dominant review, nine clusters provide limited information. The two-way calculation clusters by security identifier, not ultimate issuer, and therefore does not absorb every possible cross-share-class or corporate-group dependence.

The deletion results are more weighting-sensitive. In 2012–2016, the event-weighted mean is -2.47%, compared with an equal-review mean of -7.27%; one review contains 41.3% of the events. The jackknife standard error is 4.94 percentage points, compared with

CR1's 3.50. The differing estimates describe different weighting targets and influential cluster composition. We do not choose the weight that yields the preferred economic narrative. Appendix B reports the later-period comparison in a compact table, and the audit files retain all periods.

7.3 Returns through twenty trading days after effectiveness

Table 8 reports the new fixed-cohort extension. For additions in 2017–2026, the mean return before ED is 2.43% and the non-overlapping ED-through-ED+20 component is 0.87%. Their sum is 3.30%, with a 95% analytical interval of [1.31%, 5.28%] and an independently computed wild-bootstrap p-value below 0.001. The post-ED component itself has an interval of [-0.78%, 2.51%]. The cumulative result therefore does not establish a separately positive post-event drift.

Table 8: A fixed cohort decomposition through twenty days after implementation

Type	Period	N	Before ED	After ED	Cumulative	CR1 interval	p_{WCR}
Addition	2017–2026	169	2.43	0.87	3.30	[1.31, 5.28]	< 0.001
Addition	2022–2026	87	1.77	-0.06	1.71	[0.72, 2.69]	0.010
Deletion	2017–2026	136	-0.70	-0.38	-1.07	[-3.68, 1.54]	0.414
Deletion	2022–2026	75	-0.93	-0.95	-1.88	[-5.72, 1.96]	0.373

New exploratory audit. Before ED is CAR $[AD-1, ED-1]$; After ED is CAR $[ED, ED+20]$; Cumulative is their event-level sum, CAR $[AD-1, ED+20]$. Every entry in a row uses exactly the same complete-case events. All values are percentages. The interval is analytical CR1 with t_{G-1}; p_{WCR} is a separately computed restricted wild cluster bootstrap p-value using 49,999 draws and the finite-simulation correction $(k+1)/(B+1)$. These are distinct inferential summaries. No claim of permanent or causal price impact follows from a nonzero finite-horizon cumulative return. Source: audit/results/paired_horizon.csv.

The fixed cohort is essential to interpreting this decomposition. It ensures that the two components add at the event level and in their reported means. It does not turn the before-ED component into an estimate of a treatment effect, nor does it eliminate unrelated information arriving during the extended window. The economic content is that, for the measured later addition cohort and this benchmark, average cumulative performance has not returned to zero by the chosen endpoint. That statement concerns a finite horizon. It does not demonstrate a permanent price change.

The latest block illustrates how a longer window can change a statistical summary. Its cumulative mean is 1.71%, with an analytical interval of [0.72%, 2.69%] and wild-bootstrap $p = 0.010$. The post-ED mean is approximately -0.06%. This estimate is nevertheless sensitive to the same observations that matter in the short-window analysis. Removing Sivers lowers the cumulative mean to 1.25%, with an interval reaching below zero; removing MSCI overlaps gives 1.21%, also with an interval spanning zero. The full latest-block estimate must not be promoted as a robust finding merely because one extended-window test is significant.

The later-decade cumulative mean is less sensitive in those particular checks. It is 3.07% without Sivers, 2.51% without MSCI overlaps and 3.14% without the screened company-news observations; all three pointwise analytical intervals remain above zero.

Appendix D gives the estimates and bootstrap tests. These exclusions were added after inspecting the extension and are explicitly exploratory. Positive results from several overlapping specifications are not several independent discoveries.

For deletions, the later-decade cumulative estimate is -1.07%, with an interval of [-3.68%, 1.54%]. The latest-block value is -1.88%, also imprecise. Hence the extension does not support a symmetric addition-deletion persistence claim. Nor can the imprecise deletion estimates rule out an economically relevant negative response. The strongest defensible addition to the thesis is the transparent horizon accounting and its sensitivity analysis, rather than a new claim that Swedish indexing has a permanent causal effect.

7.4 What the auxiliary designs can resolve

The dual-share-class exercise compares an indexed security with another class from the same firm. It can remove some firm-level information, but the comparison class may be affected by the event through substitution or shared investor attention. The available set contains 39 events from 13 firms. The primary sign-adjusted difference is 0.33 percentage points on an event-weighted basis and 0.68 on a firm-weighted basis; the respective placebo p-values are 0.407 and 0.322. Neither estimates a precisely zero effect. Excluding events with price-source changes leaves 29 events from nine firms and a mean of 0.04 percentage points.

The placebo construction draws alternative seasonal windows rather than exploiting random index assignment. Its event-wise draws do not fully preserve the joint history of repeated events within a firm. That limits the interpretation of its reference distribution. A short announcement-window difference of 0.70 percentage points has an unadjusted placebo p-value of 0.029, but the archived three-window adjustment raises it to 0.088. We do not elevate that isolated result above the primary window or use the imprecise primary contrast to conclude that demand has no effect.

The inherited share-issuance exercise and OMXS30 comparison are reported in Appendix E. They are informative about the available data but do not strengthen identification sufficiently to become main results. Share-count adjustments remain partly heuristic, and the small OMXS30 sample cannot establish equivalence to OMXSB. Keeping these exercises visible while limiting their argumentative weight avoids treating every available regression as corroboration.

8 Discussion and limitations

8.1 The joint reading of prices and quantities

The central empirical pattern has three parts. Event-window addition returns are substantially lower in the latest block than in the unusually high first block. A positive average remains visible when the later decade is pooled. At the same time, trading becomes more concentrated immediately before implementation and covered benchmark-linked funds change their holdings in the index-directed direction. These facts can coexist because

trading quantities, price movements and the timing of price adjustment are distinct objects.

A market with more predictable benchmark demand and more responsive liquidity supply could display this combination. So could a market in which part of the measured return reflects selection and information while predictable trading is executed with limited additional price movement. The current evidence does not distinguish these accounts sharply enough to rank their causal importance. The appropriate inference is a constraint on simple stories: low implementation-window returns do not establish negligible index-related activity, while large holdings changes do not establish a large causal price response.

The comparison with Greenwood and Sammon (2025) is therefore most useful as a common empirical question. Sweden differs in index rules, market composition, fund coverage and the observed historical interval. The Swedish early sample is small and does not extend to the decades with the largest US estimates. A numerical difference between countries cannot be attributed to passive ownership, liquidity or arbitrage capacity without harmonized measurement and a design for that comparison. This study contributes evidence on a different setting rather than a rejection or confirmation of a universal index-effect trajectory.

The ratio evidence illustrates a broader measurement issue. If a historical ownership database captures a growing fraction of the relevant investors, its measured demand can increase even without a commensurate increase in total demand. A falling return-to-measured-demand ratio can then reflect coverage as well as price responsiveness. This is not a reason to abandon holdings data. It is a reason to report common samples, mandate definitions, report dates, denominator uncertainty and independent reporting comparisons alongside the ratio itself.

8.2 Threats to interpretation

The first limitation is identification. Reconstitution is related to size, trading and industry composition. Market adjustment and observed controls do not remove all reasons that selected securities might have different returns. The current comparison-security construction is not a reconstructed threshold experiment, and the share-class exercise can contain spillovers. Consequently, the paper estimates return patterns around events and conditional associations, rather than treatment effects of index assignment or structural stock-demand elasticities.

The second limitation is historical measurement. The event inventory is larger than the usable return panel, and a portion of the notice URLs could not be recovered during the independent retrieval audit. A successfully retrieved page is not by itself proof that every event field has been independently reconciled. Fund holdings depend on report completeness, mandate classification and historical share-count adjustments. Checksums and code replication secure the identity and transformation of the supplied files; they do not authenticate the economic content of each vendor observation.

The third limitation is statistical resolution. The final block has only nine reviews per event type, early holdings estimates have still fewer, and events can reappear across reviews. Alternative dependence estimators and review influence checks help reveal sensitivity but do not provide exact finite-sample coverage. The latest return intervals are wide relative to

economically plausible effects. An insignificant coefficient is therefore a weak basis for saying that an effect is absent, particularly when equivalence is not established.

The fourth limitation is the history of analytical choices. The repository contains several specifications, screens and previously explored breakpoints. The new audit has access to those findings. This study cannot claim that its full hypothesis family, periodization or extension was fixed independently of outcomes. The ten-test Holm calculation is a bounded sensitivity exercise, not a comprehensive correction for that history. The longer-horizon result is openly exploratory and should motivate future validation rather than be presented as a pre-specified discovery.

Finally, the code audit identifies numerical rather than purely economic uncertainty in some auxiliary intervals. Re-estimation reproduces the point estimates and sample counts, but does not reproduce every bootstrap endpoint identically. The largest discrepancies occur in sparse auxiliary ratio designs. The manuscript avoids making substantive claims from those endpoints and reports analytical and Fieller alternatives with their own assumptions. This separation matters: successful execution of a pipeline is weaker than validation of every inferential procedure it implements.

8.3 A focused route to stronger evidence

The most valuable next step is to improve the link between the observed event and a credible counterfactual, rather than to add more significance tests. A historical reconstruction of the Nasdaq ranking and eligibility rules could identify whether a defensible local comparison exists. Such a design would require the actual selection inputs at each review, the relevant industry constraints and buffers, an analysis of manipulation or sorting, and evidence that observations near any boundary are comparable. A market-capitalization screen alone is insufficient.

A second priority is to reconcile the holdings measurement at the report and share-class level. The decisive records are complete pre- and post-event reports, dated mandates, corporate-action adjustments and the exact shares-outstanding denominator. A stable-coverage panel would permit a more credible comparison of measured demand over time. Separate analyses using strict followers and broader benchmark-linked investors could then quantify the sensitivity to the investor definition without conflating it with changing event samples.

A third priority is prospective validation. Future OMXSB reviews could be analyzed under a dated protocol specifying the sample, windows, benchmark, exclusion rules and primary tests before their returns are observed. The current 2026 block is incomplete, so further events are natural additions to a new validation sample. That would test whether the later-decade patterns persist without retroactively treating the existing exploratory results as confirmatory. Until then, the present study's value rests on transparent measurement, reproducible arithmetic and appropriately bounded conclusions.

9 Conclusion

This study examines OMX Stockholm Benchmark reconstitutions from July 2007 through June 2026 using returns, reported fund holdings and trading data. The main sample contains 256 additions and 215 deletions. The average addition CAR falls from 8.52% in the first block to 1.50% in the latest block, but the intermediate periods do not form a monotonic decline. The latest estimate is imprecise, and a failure to reject zero does not establish that event returns have become economically negligible.

The later decade retains a positive addition association of 2.15%, including under several separately reported exclusions. Benchmark-linked holdings changes and increasingly concentrated trading before effectiveness show that the price pattern occurs alongside measurable investor activity. They do not identify how much of the return is caused by index demand. Early holdings ratios are particularly sensitive to a small and uncertain denominator, while later ratios remain descriptive because coverage and non-index trading are not fully resolved.

An exploratory calculation on fixed, non-overlapping return cohorts finds a positive average addition CAR through twenty trading days after effectiveness in the later decade. The result improves the accounting of the return horizon, but establishes neither a permanent price shift nor causality. Deletions and the latest addition block are less precisely resolved. The defensible conclusion is that Swedish reconstitutions exhibit changing and heterogeneous return associations whose interpretation depends on timing, measurement and inference. A complete disappearance of the effect, a structural demand elasticity and a causal mechanism remain beyond what this design can establish.

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A Definitions and reconstruction

The unit of the main analysis is a security-reconstitution event. An issuer can contribute several observations, and different share classes can have different index status. A review cycle defines the primary inference cluster. AD and ED are dates mapped to the observed trading calendar. "Latest" means the 2022–2026 block through the first 2026 review, not a complete five-year realization. The earlier and later decades are 2007–2016 and 2017–2026, respectively.

Returns are daily percentages, and reported CARs sum daily stock returns less OMXSBGI returns. A 2.15% CAR therefore denotes 2.15 percentage points in the arithmetic cumulative excess-return measure, rather than a claim that a compounded investment outperformed by exactly 2.15%. Announcement, implementation and primary windows are $[AD-1, AD+1]$, $[ED-1, ED+1]$ and $[AD-1, ED+1]$. The exploratory extension uses $[AD-1, ED-1]$ and $[ED, ED+20]$ on identical complete cases.

For an event-weighted mean $\widehat{\mu} = N^{-1}\sum_i y_i$, the independent CR1 variance estimator is

$$\widehat{V}(\widehat{\mu}) = \frac{G}{G-1}\frac{1}{N^2}\sum_{g=1}^{G}\left[\sum_{i\in g}(y_i-\widehat{\mu})\right]^2. \tag{7}$$

The intercept-only regression has one estimated parameter, so the usual $(N-1)/(N-K)$ correction equals one. The joint covariance of the return and ownership means uses the corresponding outer products of the two-dimensional review scores. The Fieller calculation solves the resulting quadratic inequality rather than dividing separate marginal confidence limits.

The cluster jackknife variance is $(G-1)/G$ times the sum of squared differences between the leave-one-review-out estimates and the full estimate. The two-way variance combines separately corrected review and security components and subtracts the corrected review-security intersection. These definitions are supplied to make the calculations inspectable; their validity still depends on the dependence structure and relevant asymptotics.

B Weighting and dependence diagnostics

The event and review means differ because review sizes vary. Neither weighting scheme is inherently more causal. They answer, respectively, what return the average observed event experienced and what return an average observed review delivered after first averaging its events. The leave-one-out range describes influence, while the standard errors summarize different maintained dependence assumptions. The complete four-block diagnostics are preserved in `audit/results/dependence_sensitivity.csv`.

Table A1: Weighting and dependence sensitivity

Type	Period	Event	Review	Leave one out	CR1	CV3	Two way
Addition	2017–2026	2.15	2.50	[1.87, 2.47]	0.55	0.55	0.47
Addition	2022–2026	1.50	1.50	[1.02, 2.01]	0.74	0.77	0.71
Deletion	2017–2026	-1.04	-0.52	[-1.45, -0.50]	0.87	0.88	0.87
Deletion	2022–2026	-1.53	-1.32	[-2.06, -0.58]	1.12	1.17	1.11

CAR $[AD - 1, ED + 1]$ in percent. Review mean gives each review equal weight. Leave one out is the range of event-weighted estimates after removing one review at a time, not a confidence interval. CR1, CV3 and Two way report standard errors. CV3 is the cluster jackknife centered on the full-sample estimate. Two way combines review and security clustering with subtraction of the review-security intersection. It does not solve dependence between different securities across different reviews. Source: audit/results/dependence_sensitivity.csv.

C Multiplicity and equivalence

Table A2: Multiplicity in a declared ten test audit family

Type	Test	Raw p	Holm p
Addition	2007–2011	0.027	0.214
Addition	2012–2016	0.908	1.000
Addition	2017–2021	0.005	0.046
Addition	2022–2026	0.062	0.310
Addition	2022–26 minus 2007–11	0.029	0.214
Deletion	2007–2011	0.007	0.059
Deletion	2012–2016	0.522	1.000
Deletion	2017–2021	0.759	1.000
Deletion	2022–2026	0.279	1.000
Deletion	2022–26 minus 2007–11	0.039	0.235

The audit family contains eight five-year total-window means and two latest-minus-earliest contrasts. It was declared after the inherited results were known. Holm adjustment controls familywise error for this family under valid marginal tests; it cannot correct unrecorded historical specification search. It does not turn the thesis into a preregistered study. Source: audit/results/multiplicity.csv.

Only the 2017–2021 addition mean remains below 0.05 under the declared ten-test Holm adjustment. The adjustment is conditional on the validity of each underlying marginal test and concerns only this family. It does not include every holdings, mechanism, horizon or historical specification examined during the project. The absence of adjusted significance in other rows neither proves a null effect nor invalidates their point estimates as descriptive summaries.

The latest-period equivalence check uses an audit margin of ±1 percentage point. The analytical 90% interval is [0.13, 2.87] for additions and [-3.62, 0.55] for deletions. Neither is contained within the margin. The archived approximate MDE for the latest addition mean is 2.35 percentage points and for the later-decade addition mean is 1.62. The last-two-period contrast has an approximate MDE of 3.06 percentage points, and the decade contrast 5.23. These plug-in quantities explain why economically relevant differences can remain

unresolved; they are not exact power calculations for the bootstrap or independent evidence of an effect.

D Exploratory horizon exclusions

Table A3: Exclusions in the exploratory cumulative return extension

Period	Sample	N	CAR	CR1 interval	p_{WCR}
2017–2026	full	169	3.30	[1.31, 5.28]	< 0.001
2017–2026	without Sivers	168	3.07	[0.96, 5.18]	0.003
2017–2026	without MSCI overlap	157	2.51	[0.60, 4.42]	0.008
2017–2026	without screened company news	163	3.14	[1.13, 5.15]	< 0.001
2022–2026	full	87	1.71	[0.72, 2.69]	0.010
2022–2026	without Sivers	86	1.25	[-0.14, 2.63]	0.087
2022–2026	without MSCI overlap	82	1.21	[-0.25, 2.67]	0.114
2022–2026	without screened company news	84	1.79	[1.04, 2.53]	0.003

CAR $[AD-1, ED+20]$ in percent, constructed on complete non-overlapping before/after cohorts. Each exclusion is separate. These follow-up diagnostics were added after inspecting the new cumulative estimates, and inherit the limitations of the original screens. Analytical intervals and 49,999-draw wild-bootstrap p-values have the same definitions as Table 8. Source: audit/results/horizon_sensitivity.csv.

The follow-up exclusions are intentionally retained alongside the complete-cohort result. Each changes the population rather than "correcting" the baseline into an uniquely valid estimate. In particular, the outcome-informed news screen is incomplete, and MSCI exclusion does not identify the incremental effect of OMXSB membership. The complete output also reports deletion estimates and all four historical blocks, preventing the presentation of the later addition result as the only horizon analyzed.

E Auxiliary evidence and its limits

The issuance exercise relates reconstitution status to the change in shares outstanding from AD-5 to ED+60, after the inherited split adjustment. Its continuous estimate is 0.57 percentage points, with an interval of [-0.47, 1.65] and $p = 0.284$. The indicator for an increase exceeding 1% has an estimated difference of 5.77 percentage points and nominal $p = 0.042$; the 5% threshold gives 3.67 percentage points and $p = 0.091$. These thresholds are part of the exploratory record. The adjustment flags 48 of 2,586 rows as split-like, and 23 changes exceeding 30 log percentage points remain. Without reconciliation against all relevant corporate actions, these results do not establish an issuer supply response to index demand. Source: `other_tests/issuance_omxs30_results.csv` and the archived construction code.

The OMXS30 exercise contains 36 usable events, 19 additions and 17 deletions, across 15 reviews in the pooled sample. The addition mean is 0.21%, with an interval of [-2.59, 3.27]; the sign-adjusted deletion mean is 0.69%, with an interval of [-3.35, 7.16]. The

archived approximate MDE for the pooled OMXS30-minus-OMXSB contrast is 3.67 percentage points. The results are too imprecise to demonstrate that the indices have equal effects or that either effect is economically zero. They also concern different selection rules and firms.

The turnover-rank exercise compares the characteristics of additions over time. Their absolute average rank changes from approximately 53.0 to 43.1; the inherited contrast has $p = 0.261$. Percentile ranks move more because the observed universe expands from roughly 316 to 709 securities. Moreover, the inherited ranking uses six months of turnover near the pre-announcement month, while the current official procedure uses monthly rankings over twelve months. It cannot be interpreted as a validated forecast of the actual Nasdaq selection rule. Source: `liquidity/liquidity_results.csv` and the size-rank aggregate files.

The share-class analysis uses sign-adjusted relative log returns, unlike the arithmetic market-adjusted CAR in the main tables. Its event-weighted and firm-weighted summaries should therefore not be mechanically subtracted from the main return means. The placebo and source-change sensitivities are retained in the aggregate twin files, including outcomes that do not support a precise relative-price response.

F Reproducibility and unresolved verification

The manuscript is tied to repository commit `d2d41666ed21`; the complete identifier is recorded in the accompanying manifest. The original analysis was rerun in a separate directory. Its full runner completed with zero reported execution failures, and the inherited input checks passed for 43 files. An independently written audit reconstructed all three inherited CAR windows for 487 candidates and passed 73 recorded checks. The checks cover sample counts, arithmetic, date mapping, duplicate keys and comparisons of selected estimators. They are neither 73 independent confirmations of the economic hypothesis nor a certification of every source observation.

A comparison of 40 numerical CSV outputs finds 32 matching within the stated 10^{-7} tolerance. Eight contain differences in bootstrap intervals or simulated p-values. Point estimates, counts and standard errors agree within the audit tolerances. The maximum absolute difference in the main effect table is 0.0043, while some sparse auxiliary composition-ratio intervals differ substantially. One early addition ratio interval changes from [-52.03, 417.95] in the frozen output to [-25.21, 96.37] in the rerun. Another changes from an unbounded interval to a bounded one. Those unstable endpoints are not used to substantiate manuscript conclusions.

The rerun uses Python 3.12.4, NumPy 1.26.4, pandas 2.2.3 and SciPy 1.15.2. These differ from the environment recorded by the inherited project. The bootstrap routines use finite simulation and numerical test inversion; the observed differences are consistent with sensitivity to their numerical implementation, but the audit does not establish a single exclusive cause. The original code and results remain preserved. The new extension, Fieller calculations, checks and generated exhibits are provided separately.

Initial external retrieval recovered 23 of the 39 review notices. Follow-up research

opened notices or alternative original attachments for another 12 reviews; three additional reviews were visible only through indexed text, and the 2008 second-half notice remained unresolved. Some accessible documents are secondary republications. Retrieval is separate from complete field-level reconciliation, so these counts do not certify 35 fully verified reviews. A December 2008 correction notice also prints an internally inconsistent effective year; the dataset uses January 2009, but the timing of the initial announcement still requires reconciliation. Original Nasdaq methodology documents and three example thesis PDFs were recovered. Research references were checked against primary publication records or available primary texts, with full-text availability recorded separately.

The accompanying package contains the manuscript, aggregate exhibit inputs, original analysis code, independent audit code, manifests, source links and an issue register. It does not newly redistribute licensed security-level vendor data or third-party full texts. A researcher with lawful access must supply the documented inputs to rerun the complete empirical chain. Checksums identify the version used but are not a substitute for access rights. The authors must still resolve notice discrepancies, reconcile relevant holdings reports and corporate actions, and make the permitted-use and submission decisions themselves.

G Writing benchmark and AI disclosure

The organizational benchmark is the BE451 Autumn 2026 syllabus and introductory research material, together with past SSE finance theses available in the school's archive (Stockholm School of Economics, 2026). The syllabus specifies a maximum of 40 single-spaced, 12-point pages excluding appendices; its guidance places the main text at approximately 25–40 pages from introduction to conclusion, including major exhibits, and recommends no table of contents. This draft follows an empirical-paper structure: question and contribution, focused literature, institutional setting and data, methods, results, mechanisms, robustness, discussion and conclusion. This is an application of documented course guidance, not a claim to know the examiner's private preferences or an award committee's future ranking.

The inspected examples include Fager and Öhman Tunklint (2024), Holst and von der Esch (2023) and Domeij and Krieg (2021). Their public archive identifiers are 6053, 5734 and 5028. They inform the emphasis on a narrow question, self-contained exhibits and a visible boundary between design and interpretation. No empirical findings or prose are copied from those theses. The accompanying structure note identifies the examples and documented course requirements.

AI disclosure for this working document. OpenAI Codex assisted with source retrieval, computations, code review, exhibits, layout and manuscript composition. The English prose was AI-generated. The course syllabus permits disclosed assistance but prohibits AI directly writing thesis text. This is therefore a research aid, not a compliant submission manuscript. The authors must write the submitted text, verify its argument and sources, disclose assistance accurately and take responsibility for the work.